# fala versio

## _travelers guide
## _for speaking to aliens

| 24 | 3 | 7 |
|---|---|---|
| bandpass filters | formants | knobs · 2 switches · tap |

[ tap ] press  [ tap ] hold  › then  noon = 12 o'clock  numbers = the numbered panel

stereo formant filter · up to 24 bandpass filters · clockable lfo · a free firmware for any versio

← back to the versio starsystem

purple = hold

## 01 what it is

fv_01

| | |
|---|---|
| a formant filter | stereo vowel filter · up to 24 bandpass filters, a wavefolder before, saturation after |
| name | portuguese for "speak" |
| in 1 | audio in · a saw or other rich wave works best |
| out 1 out 2 | stereo out |
| swap in | free on any versio · how to swap |

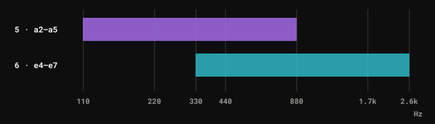
## 02 the formants

fv_02 · knobs 5 · 6 · 3 · fig.01

| | |
|---|---|
| 5 [ a2–a5 ] | formant a · 110 – 880 Hz, three octaves |
| 6 [ e4–e7 ] | formant e · 330 – 2637 Hz · silent with abc on a |
| 3 [ q ] | resonance of every filter at once |
| musical | ranges follow notes, not voices, so each knob spans whole octaves |

### __fig.01 : formant ranges

whole octaves · knobs 5 + 6




## 10 first flight

fv_10 · the manual's first patch

1. a saw into in 1 · out 1 out 2 to your mixer
2. 1 [ fold ] + 7 [ saturation ] min · 4 [ index ] noon
3. abc ae (mid)
4. sweep 5 [ a2–a5 ] and 6 [ e4–e7 ] to talk
5. raise 3 [ q ] for sharper vowels
6. then index for movement · fold and saturation for grit

## 03 fold · saturate

fv_03 · knobs 1 + 7

| | |
|---|---|
| 1 [ fold ] | wavefolding before the filters |
| 7 [ saturation ] | drive after the filters |
| levels | a built-in vca keeps the volume even as you move things |

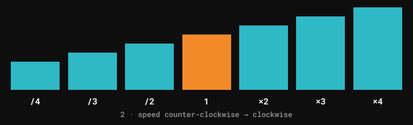
## 04 lfo

fv_04 · knobs 2 + 4 · fig.02

| | |
|---|---|
| 2 [ speed ] | lfo rate |
| clocked | speed becomes /4 /3 /2 1 ×2 ×3 ×4 of the clock |
| 4 [ index ] | stereo lfo depth on the filters · noon off |
| left · right | left smooth random · right sine |

### __fig.02 : speed when clocked

knob 2 · seven ratios




### __fig.04 : signal path

as the manual describes it




the stereo lfo (2 speed · 4 index) moves the formants

## 05 switches

fv_05 · abc · xyz

| | |
|---|---|
| abc left | [ a ] formant a only |
| abc mid | [ ae ] formants a + e |
| abc right | [ ae3 ] adds a third, worked out from a and e |
| xyz | [ i ] · [ ii ] · [ iv ] one, two or four parallel sets · chorus-like |
| all in | ae3 + iv = 24 bandpass filters, 12 saturators |

## 07 tap · clock

fv_07 · fsu = tap

| | |
|---|---|
| [ tap ] | tap a tempo for the lfo |
| fsu jack | clock or trigger pattern in |
| [ tap ] few s | forget the tempo · lfo runs free again |
| boot | cyan · magenta · cyan · magenta = fala is loaded |

## 08 five patches

fv_08 · from the manual · abc · xyz

**basic filter** — ae · any
fold + saturation min · index noon · a saw into in 1 · move a, e and q

**animated** — any
index up · speed, or tap a tempo

**grimy** — any
try the switches · raise fold and saturation

**rhythmic sync** — any
a drum loop in · a copy of its clock into the tap jack

**feedback stereo** — any
mono source into in 1 · mult out 1 back into in 2 · listen to both outs (after starthief)

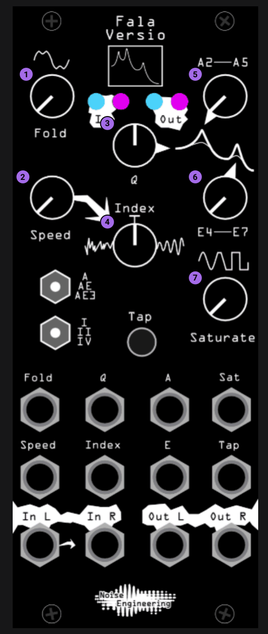
### __fig.06 : what the LEDs say

at boot

boot — fala is loaded

### __fig.03 : the panel

numbers match the legend




1 fold · wavefolder in
2 speed · lfo rate · or ratio
3 q · resonance
4 index · random ← · → sine
5 a2–a5 · 110 – 880 Hz
6 e4–e7 · 330 – 2637 Hz
7 saturation · drive out
abc formants · a · ae · ae3
xyz sets · i · ii · iv
fsu tap · tap · clock in
in in 1 · audio in
out out 1 / 2 · stereo out

## 09 specs

fv_09

| | |
|---|---|
| cv in | 0 – 5 V |
| size | 10 hp · 1.5 in deep |
| power | version a 70 mA +12 V · 70 mA –12 V · version b 125 mA · 10 mA · no 5 V |